

Integrative Analysis of ChIP-Chip and ChIP-Seq Dataset

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Abstract

Epigenetic regulation and interactions between transcription factors and regulatory genomic regions play crucial roles in controlling transcriptional regulatory networks that drive development, environmental responses, and disease. Chromatin immunoprecipitation (ChIP) followed by high-throughput sequencing (ChIP-seq) and ChIP followed by genomic tiling microarray hybridization (ChIP-chip) are the two of the most widely used technologies for genome-wide identification of DNA protein interactions and histone modification *in vivo*. Many algorithms and tools have been developed and evaluated that allow identification of transcription factor binding sites from ChIP-seq or ChIP-chip datasets. However, binding site identification is only the first step; the ultimate goal is to discover the regulatory network of the transcription factor (TF). Here, we present a common workflow for downstream analysis of ChIP-chip and ChIP-seq with an emphasis on annotating binding sites and integration with gene expression data to identify direct and indirect targets of the TF. These tools will help with the overall goal of unraveling transcriptional regulatory networks using datasets publicly available in GEO.

Key words ChIP-chip, ChIP-seq, Gene expression, Regulatory network, Microarray, GO enrichment analysis, Pathway analysis, Integrated analysis, Transcription factor, Motif discovery

1 Introduction

One of the most fundamental questions in biology is how the regulation of gene expression patterns is orchestrated so precisely for different cell types, different tissues, and different developmental stages, and what goes wrong during the development of various diseases. The basic building block of the regulatory network is the site where transcription factors (TF) bind to DNA, turning a gene on or off. One of the approaches to untangle the complex regulatory network of a TF is to discover the binding sites of a particular TF in the genome and determine whether TF binding affects the expression of nearby genes. Chromatin immunoprecipitation (ChIP) followed by high-throughput sequencing (ChIP-seq) and ChIP followed by genomic tiling microarray hybridization (ChIP-chip) have become standard high-throughput technologies for genome-wide identification of TF target sites and histone modification *in vivo* [1–4].

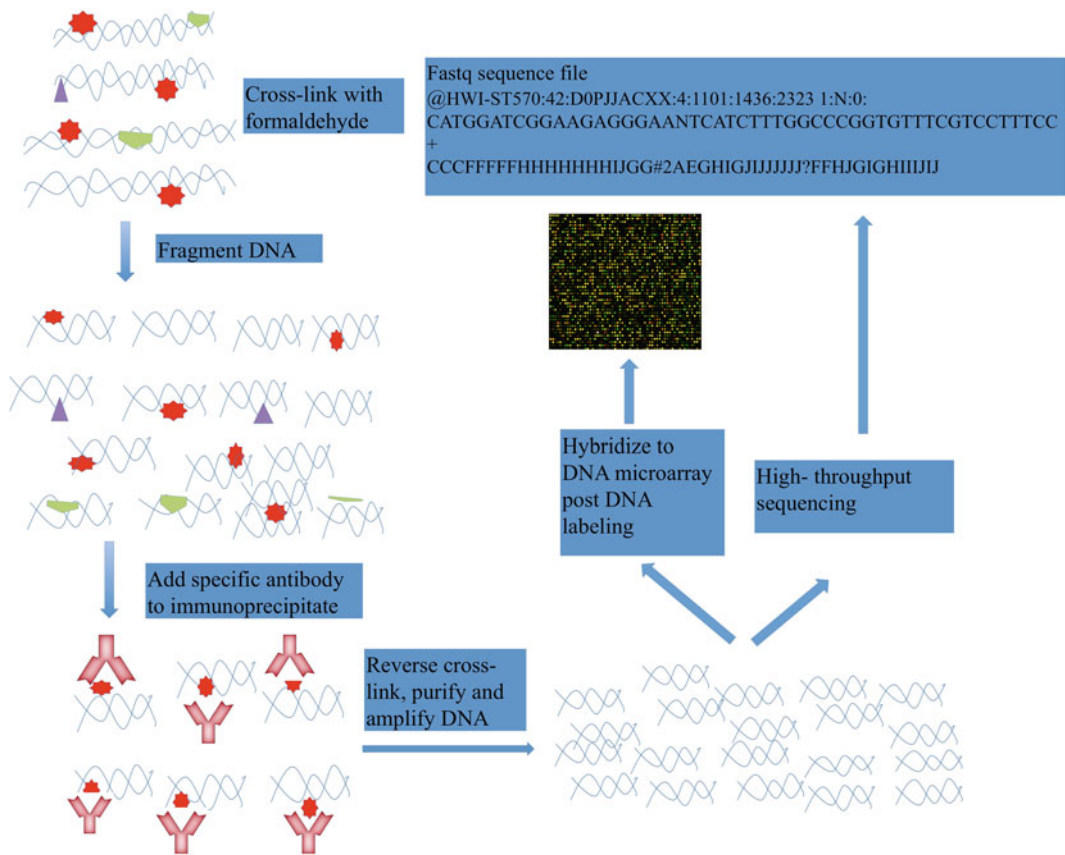


Fig. 1 Wet-lab workflow for ChIP-seq and ChIP-chip experiments. DNA and proteins are cross-linked and purified with an antibody specific to the protein of interest. Purified DNA is amplified followed by high-throughput sequencing or by hybridization to tiling Microarrays

Proteins and DNA are cross-linked with formaldehyde, followed by fragmentation via sonication and immunoprecipitation with an antibody specific to a TF. The immunoprecipitated DNA fragments are subject to sequencing or hybridization to a genomic tiling microarray (Fig. 1). Many technical considerations to obtain high-quality data are common to ChIP-seq and ChIP-chip, including antibody selection, fragment size control, amplification bias, appropriate controls, and sufficient numbers of biological replicates (reviewed in refs. 5 and 6). For a complete comparison of ChIP-chip and ChIP-seq methods, please see previous reviews [7, 8]. The effects of different tiling array platforms, amplification protocols, and signal detection algorithms have been compared using a ChIP-chip experiment with predefined DNA targets [4]. For a technical review of template preparation, sequencing and imaging, genome alignment and assembly approaches, and current NSG platforms, please see the review in ref. 9.

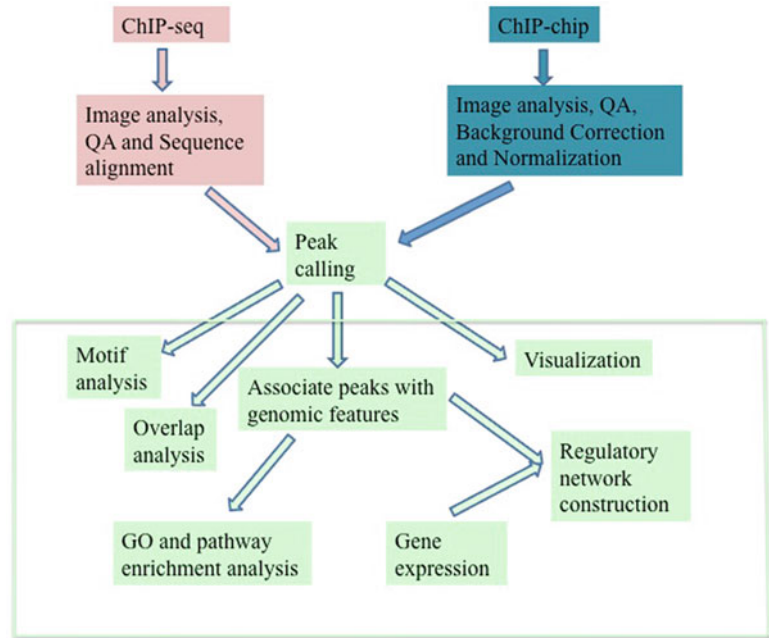


Fig. 2 Analysis workflows for ChIP-seq and ChIP-chip experiments. ChIP-seq analysis workflow and ChIP-chip workflow share a common downstream analysis workflow as depicted inside the *green box*, including associating peaks to genomic features, overlap analysis, motif discovery, GO and pathway enrichment analysis, and regulatory network construction (Color figure online)

Figure 2 depicts a typical analysis workflow for ChIP-seq and ChIP-chip. For ChIP-seq experiments, sequence information is extracted from image files with image analysis software such as Illumina Firecrest and Bustard applications. Sequences are typically saved in a FASTQ format [10]. Although several variants of FASTQ format exist differing in the quality score definition and header annotation, each sequence is represented in FASTQ file in four lines in the order of header line starting with @, sequence line, signal end of sequence line starting with +, and read base quality score (see example in Fig. 1). The quality score and the header annotation information are frequently used as quality assessment (QA) and read quality filters to remove low confidence sequencing reads. The remaining high-quality sequences are mapped onto the reference genome using software such as Bowtie [11], BWA [12], and SOAP [13] (reviewed in ref. 14). A peak-calling algorithm, such as MACS [15], FindPeaks [16], GeneTrack [17], QuEST [3], SISSRs [18], Useq [19], BayesPeak [20], CisGenome [21], CSAR [22], DIME [23], and PICS [24], is applied to identify the binding sites (also referred to as peaks) that are significantly enriched in the ChIP sample compared to the control sample and/or enriched in the local region compared to the immediate neighboring

region(s), sometimes taking advantage of strand-specific density profiles, and/or genome-wide read mappability. In general, these peak detection algorithms can be divided into sliding window approaches such as CisGenome, MACS, and Useq; local maxima estimation approaches such as FindPeaks and PICS; and hidden Markov Model (HMM) approaches such as BayesPeak. For a detailed evaluation and comparisons among these algorithms, please see reviews in ref. 25, 26. Peak locations are usually converted to a format, such as BED or Wiggle (WIG), that can be uploaded to the UCSC Genome Browser, an open-access, web-based, up-to-date source for genome sequence data integrated with a large collection of related annotations [27]. Batch annotation tools, such as ChIPpeakAnno [28], are commonly used to associate peaks to the rich genomic annotations, such as promoter region, miRNA, exon, intron, CpG island, gene ontology, and KEGG pathway. In addition, peak sequences are frequently extracted for de novo discovery of TF DNA binding site motifs using tools such as MEME [29], DREME [30], and rGADEM [31].

Overall, a ChIP-chip data analysis workflow is very similar to that of ChIP-seq. The unique aspect of ChIP-chip data analysis involves background correction and signal intensity normalization within array and among arrays prior to peak calling (reviewed in ref. 4). The subsequent downstream analysis is exactly the same as in ChIP-seq.

Here we present a common downstream analysis workflow shared by ChIP-chip and ChIP-seq experiments. Step-by-step analysis of a publicly available dataset is illustrated using tools mainly developed in the R environment, including associating peaks with rich genomic annotations, analysis of GO and pathway enrichment, detection of *bona fide* TF binding motifs and construction of regulatory networks integrated with gene expression datasets. These downstream steps are shown enclosed in the green box in Fig. 2. Publicly available ChIP-chip data for the *C. elegans* TF DAF-12 and gene expression data for animals lacking this TF are used as an example and some recovered TF binding motifs, GO terms, and interaction pathways are consistent with the known functions of this protein.

2 Materials

ChIP-chip peaks were downloaded from GEO at <http://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE28350> [32] (see Note 1). Expression Microarray results were downloaded from [33]. Software tools are listed in Table 1 (see Note 2).

Table 1
Software tools for downstream analysis of ChIP-chip and ChIP-seq

Software	URL	Reference
R 2.15 or later	http://cran.r-project.org/	[38]
ChIPpeakAnno 2.5.8 or later	http://www.bioconductor.org/packages/release/bioc/html/ChIPpeakAnno.html	[28]
liftOver	http://genome.ucsc.edu/cgi-bin/hgLiftOver	[39]
rtracklayer	http://www.bioconductor.org/packages/release/bioc/html/rtracklayer.html	[40]
rGADEM	http://www.bioconductor.org/packages/release/bioc/html/rGADEM.html	[31]
MotIV	http://www.bioconductor.org/packages/release/bioc/html/MotIV.html	[36, 41]
STAMP	http://www.benoslab.pitt.edu/stamp/	[36]
GeneNetworkBuilder	http://pgfe.umassmed.edu/bioinformatics/GeneNetworkBuilder/ http://www.bioconductor.org/packages/release/bioc/html/GeneNetworkBuilder.html	[42]

3 Methods

3.1 Determining and Visualizing the Concordance Among the Biological Replicates

In the first step, load the *ChIPpeakAnno* package and three example datasets as bed format that contain DAF-12 binding sites identified in *C. elegans* from three biological replicates [32].

```
Library(ChIPpeakAnno)
DAF12.peaks.rep1 = BED2RangedData(read.
table("/Users/zhuji/Documents/Publication/
JulieZhu/BookChapter/dataset/DataForExample/
GSM700907_DAF-12_TAP_vs_INPUT_#1_WS170.bed",
sep="\t", skip=5))
DAF12.peaks.rep2 = BED2RangedData(read.
table("/Users/zhuji/Documents/Publication/
JulieZhu/BookChapter/dataset/DataForExample/
GSM700908_DAF-12_TAP_vs_INPUT_#2_WS170.bed",
sep="\t", skip=5))
DAF12.peaks.rep3 = BED2RangedData(read.
table("/Users/zhuji/Documents/Publication/
JulieZhu/BookChapter/dataset/DataForExample/
GSM700909_DAF-12_TAP_vs_INPUT_#3_WS170.bed",
sep="\t", skip=5))
```

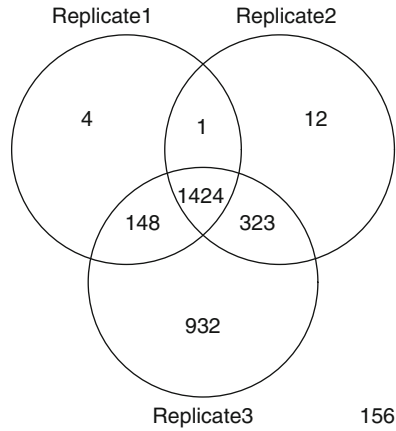


Fig. 3 Overlapping of putative DAF-12-binding sites among three biological replicates in worm. The Venn diagram was generated from the putative DAF-12-binding sites of three biological replicates in worm [32]. Hypergeometric test shows that there is significant overlapping among biological replicates (p -value $< 2e-48$ for all three pair-wise comparisons)

In the next step, generate a Venn diagram using the function *makeVennDiagram* to visualize the overlap among the three replicates (Fig. 3). In addition, pair-wise overlapping significance tests demonstrate significant concordance among replicates (p -value $< 2e-48$) (see **Note 3**).

```
pdf("Fig.3.pdf")
makeVennDiagram(RangedDataList(DAF12.peaks.
  repl, DAF12.peaks.rep2, DAF12.peaks.rep3),
  NameOfPeaks=c("Replicate1", "Replicate2", "Replicate3"),
  maxgap=0, totalTest=3000)
dev.off()
```

3.2 Combining Peaks from Replicates, Filtering Peaks Present in Negative Control and Associating the Merged Peaks with Features of Interest

Prior to associating features of interest with the peaks, it is a common practice to combine peaks from replicates.

```
MergedPeaks=findOverlappingPeaks(findOverlappingPeaks(DAF12.peaks.repl, DAF12.peaks.rep2,
  maxgap=0, select="first", NameOfPeaks1="R1", NameOfPeaks2="R2")$MergedPeaks, DAF12.peaks.
  rep3, maxgap=0, select="first", NameOfPeaks1="Replicate1Repliate2", NameOfPeaks2="R3")$Merg
  edPeaks
```

Then any peaks also present in the DAF-12(0) negative control are filtered out.

```
negative.control=BED2RangedData(read.table("/Users/zhuji/Documents/Publication/
  JulieZhu/BookChapter/dataset/DataForExample/GSM700906_AA86_vs_INPUT_WS170.bed", sep="\t",
  skip=5))
```

Fig. 4 A screenshot of the liftover tool in the UCSC Genome Browser. The UCSC liftover tool was used to convert genome coordinates from WS170 to WS220

```
final.peaks=MergedPeaks[!rownames(MergedPeaks) %in% rownames(findOverlappingPeaks(MergedPeaks, negative.control, select="first", maxgap=0, NameOfPeaks1="M", NameOfPeaks2="NC"))$Peaks1withOverlaps),]
```

The downloaded peaks are from an old genome version WS170. However, the current genome version of *C. elegans* is WS220. To convert WS170 coordinates to corresponding WS220, save the final peaks as a file in a six-field BED format (*see Note 4*) and use the liftover tool in UCSC Genome Browser illustrated in Fig. 4. Alternatively, the *liftOver* function implemented in the *rtracklayer* package can be used instead.

```
temp=as.data.frame(final.peaks)[,-4]
temp[,1]=paste("chr", temp[,1], sep="")
temp[,5]=0
temp[,6]="+"
write.table(temp, file="ws170peaks.txt", sep="\t", row.names=FALSE, col.names=FALSE, quote=FALSE)
WS220.peaks=BED2RangedData(read.table("WS220.bed", sep="\t", header=FALSE))
```

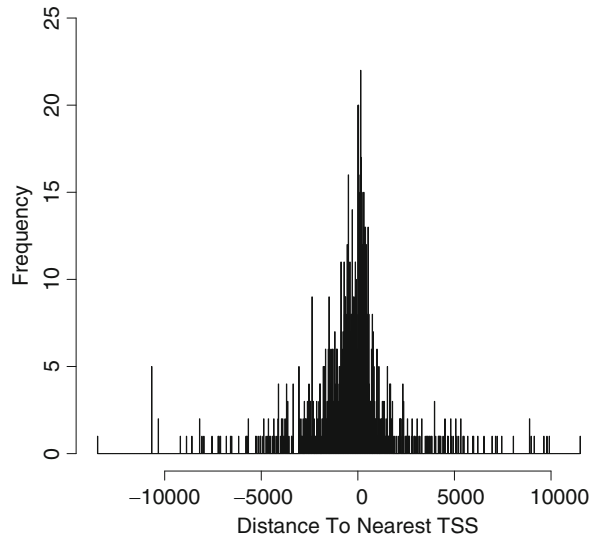



Fig. 5 Distribution of DAF-12-binding sites relative to nearest TSSs in worm. The histogram was generated from the putative DAF-12-binding regions merged from three biological replicates identified in worm [32]. The plot shows that DAF-12-binding sites are enriched in regions more symmetrically around transcription start sites. The mean of the distance to the nearest TSS is -330 ± 62 bases (mean $\pm$ SEM)

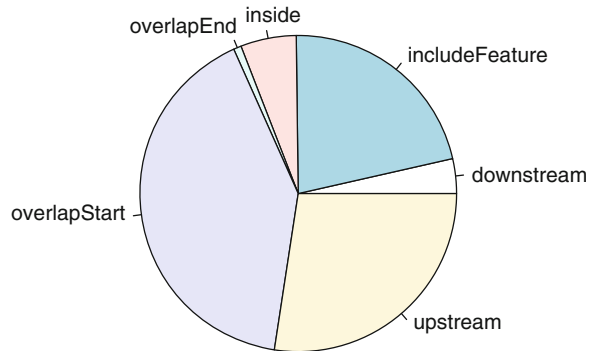


Fig. 6 Pie chart of DAF-12-binding sites relative to nearest genes in worm. The pie chart was generated from the putative DAF-12-binding regions merged from three biological replicates identified in worm [32]. The plot shows that DAF-12-binding sites are distributed more on upstream and around the genes

Associating the final peaks with the nearest gene using *annotatePeakInBatch* function reveals that DAF-12 binding sites are distributed densely around TSSs (Figs. 5 and 6).

```
mart = useMart(biomart = "ensembl", dataset =  
"celegans_gene_ensembl") (see Note 5)  
TSS = getAnnotation(mart, featureType = "TSS")  
(see Note 6)
```



```
annotated.DAF12.peaks=annotatePeakInBatch(WS220.peaks, AnnotationData=TSS) (see Note 7)
```

Additional annotations such as entrez ID, gene symbol, and gene name can be added using function *addGeneIDs* and the annotated peaks can be saved as an Excel file or plotted for visualizing the peak distribution relative to the genomic features of interest.

```
library(org.Ce.eg.db)
annotated.DAF12.peaks=addGeneIDs(annotated.DAF12.peaks,"org.Ce.eg.db",c("entrez_id",
"symbol","genename")) (see Note 8)
write.table(as.data.frame(annotated.DAF12.peaks), file="AnnotatedDAF-12BindingSites.xls",
sep="\t", row.names=FALSE)
y = annotated.DAF12.peaks$distancetoFeature[!is.na(annotated.DAF12.peaks$distancetoFeature)]
pdf("Figure5.pdf")
hist(y, xlab="Distance To Nearest TSS",
main="", breaks=1000, xlim=c(min(y)-100,
max(y)+100), ylim=c(0,25))
dev.off()
pdf("Figure6.pdf")
pie(table(annotated.DAF12.peaks$insideFeature))
dev.off()
```

3.3 Finding the Peaks That Reside in Bi-directional Promoters

The analysis result shows that more than 77 % of DAF-12 peaks reside in bi-directional promoters (*see Note 9*). It is important to include genes associated to the peaks in both directions for subsequent direct target identification and for GO and pathway enrichment analysis.

```
BDP = peaksNearBDP(WS220.peaks,
AnnotationData=TSS, MaxDistance =5000)
c(BDP$percentPeaksWithBDP, BDP$n.peaksWithBDP, BDP$n.peaks)
all.genes = union(annotated.DAF12.peaks$feature, BDP$peaksWithBDP$feature)
```

3.4 Summarizing Peak Distributions Over Exon, Intron, Enhancer, Proximal Promoter, 5' UTR and 3' UTR Regions

Obtain the annotations using *getAnnotation* followed by calling function *assignChromosomeRegion* to summarize the relative peak distributions among different chromosome regions.

```
utr5=getAnnotation(mart, featureType="5utr")
utr3=getAnnotation(mart, featureType="3utr")
exon=getAnnotation(mart, featureType="Exon")
Feature.distribution=assignChromosomeRegion(WS220.peaks, exon, TSS, utr5, utr3)
```

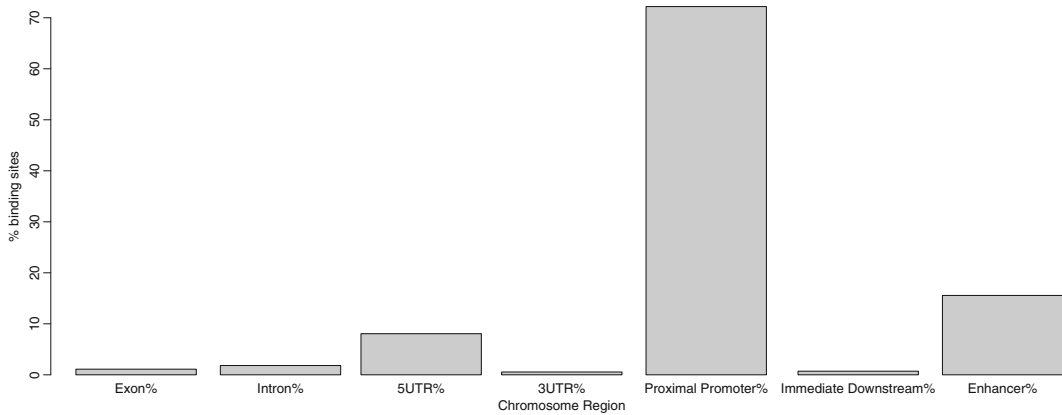


Fig. 7 Distribution of DAF-12 binding sites among various features in worm. The bar plot was generated from the putative DAF-12-binding regions merged from three biological replicates identified in worm [32]. The plot shows that 70 % DAF-12-binding sites reside in proximal promoter regions

The summarized distribution can be plotted as a bar graph and saved in pdf format.

```
pdf("Figure7.pdf")
barplot(unlist(Feature.distribution), cex.
names=1, xlab="Chromosome Region", ylab="%
binding sites", main="")
dev.off()
```

Figure 7 shows that 70 % DAF-12 binding sites reside in proximal promoter regions (*see Note 10*).

3.5 Obtaining Enriched GO Terms for Genes Near the Peaks

Load the organism-specific GO gene mapping package before acquiring enriched GO terms using *getEnrichedGO* (*see Note 11*).

```
library(org.Ce.eg.db)
enrichedGO<- getEnrichedGO (all.genes,
orgAnn="org.Ce.eg.db", maxP=0.05, multi-
Adj=TRUE, minGOTerm=10, multiAdjMethod="BH")
(see Note 8 and 11)
write.table(enrichedGO$bp, file="Table2_
EnrichedBiologicalProcesses.xls", sep="\t",
row.names=FALSE)
write.table(enrichedGO$mf, file="Table2_
EnrichedMolecularFunctions.xls", sep="\t",
row.names=FALSE)
write.table(enrichedGO$cc, file="Table2_
EnrichedCellularComponents.xls", sep="\t",
row.names=FALSE)
```

Table 2 shows a list of enriched GO molecular functions and biological processes for DAF-12 binding sites (*see Note 11*). Note

Table 2
Enriched GO molecular functions and biological processes of DAF-12-fbinding sites in worm ordered by false discovery rate (FDR)

GO ID	GO term	GO definition	Category	FDR
GO:0005515	Protein binding	Interacting selectively and non-covalently with any protein or protein complex (a complex of two or more proteins that may include other nonprotein molecules)	MF	4.85E-08
GO:0003735	Structural constituent of ribosome	The action of a molecule that contributes to the structural integrity of the ribosome	MF	0.0002
GO:0002119	Nematode larval development	The process whose specific outcome is the progression of the nematode larva over time, from its formation to the mature structure. Nematode larval development begins with the newly hatched first-stage larva (L1) and ends with the end of the last larval stage (for example the fourth larval stage (L4) in <i>C. elegans</i>). Each stage of nematode larval development is characterized by proliferation of specific cell lineages and an increase in body size without alteration of the basic body plan. Nematode larval stages are separated by molts in which each stage-specific exoskeleton, or cuticle, is shed and replaced anew	BP	0.0013
GO:0002164	Larval development	The process whose specific outcome is the progression of the larva over time, from its formation to the mature structure. The larva is the early, immature form of an animal that at birth or hatching is fundamentally unlike its parent and must metamorphose before assuming the adult characters	BP	0.0013
GO:0009791	Post-embryonic development	The process whose specific outcome is the progression of the organism over time, from the completion of embryonic development to the mature structure. See embryonic development	BP	0.0013
GO:0015078	Hydrogen ion transmembrane transporter activity	Catalysis of the transfer of hydrogen ions from one side of a membrane to the other	MF	0.0029
GO:0040034	Regulation of development, heterochronic	Any process that modulates the rate, frequency, or extent of the progression an integrated living unit: an anatomical structure (which may be a subcellular structure, cell, tissue, or organ), or organism over time from an initial condition to a later condition at a consistent predetermined time point during its development	BP	0.039

(continued)

Table 2
(continued)

GO ID	GO term	GO definition	Category	FDR
GO:0000166	Nucleotide binding	Interacting selectively and non-covalently with a nucleotide, any compound consisting of a nucleoside that is esterified with (ortho)phosphate or an oligophosphate at any hydroxyl group on the ribose or deoxyribose	MF	0.049
GO:0005488	Binding	The selective, non-covalent, often stoichiometric, interaction of a molecule with one or more specific sites on another molecule	MF	0.049
GO:0017076	Purine nucleotide binding	Interacting selectively and non-covalently with purine nucleotides, any compound consisting of a purine nucleoside esterified with (ortho)phosphate	MF	0.049
GO:0032553	Ribonucleotide binding	Interacting selectively and non-covalently with a ribonucleotide, any compound consisting of a ribonucleoside that is esterified with (ortho)phosphate or an oligophosphate at any hydroxyl group on the ribose moiety	MF	0.049
GO:0032555	Purine ribonucleotide binding	Interacting selectively and non-covalently with a purine ribonucleotide, any compound consisting of a purine ribonucleoside that is esterified with (ortho)phosphate or an oligophosphate at any hydroxyl group on the ribose moiety	MF	0.049
GO:0035091	Phosphatidylinositol binding	Interacting selectively and non-covalently with any inositol-containing glycerophospholipid, i.e. phosphatidylinositol (PtdIns) and its phosphorylated derivatives	MF	0.049
GO:0035639	Purine ribonucleoside triphosphate binding	Interacting selectively and non-covalently with a purine ribonucleoside triphosphate, a compound consisting of a purine base linked to a ribose sugar esterified with triphosphate on the sugar	MF	0.049
GO:0051015	Actin filament binding	Interacting selectively and non-covalently with an actin filament, also known as F-actin, a helical filamentous polymer of globular G-actin subunits	MF	0.049

The enriched GO terms were obtained including genes associated to the peaks in both directions within 5 kb to the TSS. The parameter used to generate the list is *maxP*=0.05, *multiAdj*=TRUE, *minGOterm*=10, and *multiAdjMethod*= "BH"

Table 3
Enriched pathways in reactome database for DAF-12 binding sites in worm ordered by FDR

Pathway ID	Pathway definition	FDR
1626134	<i>Caenorhabditis elegans</i> : Regulation of gene expression in beta cells	0.004
1626136	<i>Caenorhabditis elegans</i> : Diabetes pathways	0.004
1625991	<i>Caenorhabditis elegans</i> : Peptide chain elongation	0.004
1625992	<i>Caenorhabditis elegans</i> : Eukaryotic Translation Elongation	0.004
1625772	<i>Caenorhabditis elegans</i> : GTP hydrolysis and joining of the 60S ribosomal subunit	0.006
1626131	<i>Caenorhabditis elegans</i> : Regulation of beta-cell development	0.006
1625983	<i>Caenorhabditis elegans</i> : Eukaryotic Translation Termination	0.007
1626135	<i>Caenorhabditis elegans</i> : Insulin Synthesis and Processing	0.023
1625880	<i>Caenorhabditis elegans</i> : Developmental Biology	0.032
1625773	<i>Caenorhabditis elegans</i> : Formation of a pool of free 40S subunits	0.036

The enriched pathway terms were obtained including genes associated to the peaks in both directions within 5 kb to the TSS. The parameter used to generate the list is *maxP*=0.05, *multiAdj*=TRUE, *minGOterm*=10, and *multiAdjMethod*="BH"

that several of the terms are associated with development, consistent with the known role of DAF-12 in regulation of dauer formation in *C. elegans*.

3.6 Obtaining Enriched Pathway Terms Near the Peaks

Load the pathway database Reactome before acquiring enriched pathway terms using *getEnrichedPATH*.

```
library(reactome.db)
enrichedPATH=getEnrichedPATH(all.genes,
orgAnn="org.Ce.eg.db", pathAnn="reactome.db",
maxP=0.05, minPATHterm=10, multiAdjMethod="BH")
(see Note 12)
write.table(enrichedPATH, file="Table3_
EnrichedPATHsWithGeneIDs.xls", sep="\t", row.
names=FALSE)
```

Table 3 shows a list of enriched pathways from the Reactome database for DAF-12 binding sites (see **Note 12**). Note that the enriched insulin and diabetes pathways are associated with *C. elegans* dauer formation, consistent with the known role of DAF-12 in this process.

3.7 Retrieving the Sequences Around the Binding Sites for PCR Amplification or Motif Discovery

First, load the organism-specific *BSgenome* package followed fetching sequences using *getAllPeakSequence*.

```
library(BSgenome.Celegans.UCSC.ce10) (see
Note 13) .
```

```

    peaksWithSequences=getAllPeakSequence(WS220.
    peaks, upstream=0,
    downstream=0, genome=Celegans)
    write2FASTA(peaksWithSequences, file="DAF-
    12peaks.fa", width=50)

```

Next, load *rGADEM library* for de nova motif discovery.

```

    library(rGADEM)
    sequences<- read.DNAStringSet("DAF-12peaks.fa",
    "fasta")
    gadem<-GADEM(sequences,verbose=1,genome=Cel
    egans)
    consensus(gadem)
    motifs<- getPWM(gadem)
    motifs.trimed<- trimPWMedge(motifs, threshold=1)

```

Last, load *MotIV* library for matching discovered motifs with known motifs.

```

    library(MotIV) (see Note 2)
    DAF12.analysis.jaspar<- motifMatch(motifs.
    trimed)
    summary(DAF-12DAF12.analysis.jaspar)
    pdf("Figure8.pdf")
    plot(DAF12.analysis.jaspar,ncol=2,
    rev=FALSE, main="Logo", bysim=TRUE)
    dev.off()

```

Significant motifs (p -value<0.0002) identified by GADEM with matching motifs in JASPAR database [34] are given in Fig. 8 as sequence logos, a graphical representation of sequence alignments [35] (see **Note 14**). An alternative web tool for motif matching is STAMP [36]. Note that DAF-12 is a member of the nuclear hormone receptor family of TFs and cooperates with DAF-16, a member of the forkhead family of TFs; our analysis recovered motifs for a mammalian hormone receptor, NR2F1, and a forkhead TF, FOXD3.

3.8 Constructing Regulatory Network Combining with Expression Dataset

Intersecting genes with promoter regions bound by TAF-12 from the ChIP experiment and genes differentially expressed in TAF-12 mutants allow us to identify genes directly regulated by TAF-12. Genes differentially expressed in TAF-12 mutant but not bound by TAF-12 are predicted indirect targets of TAF-12. Figure 9, generated using GeneNetworkBuilder [42], gives a system-level view of the intertwined connections among the direct and indirect targets of DAF-12, which shows that DAF-12 is a master regulator.

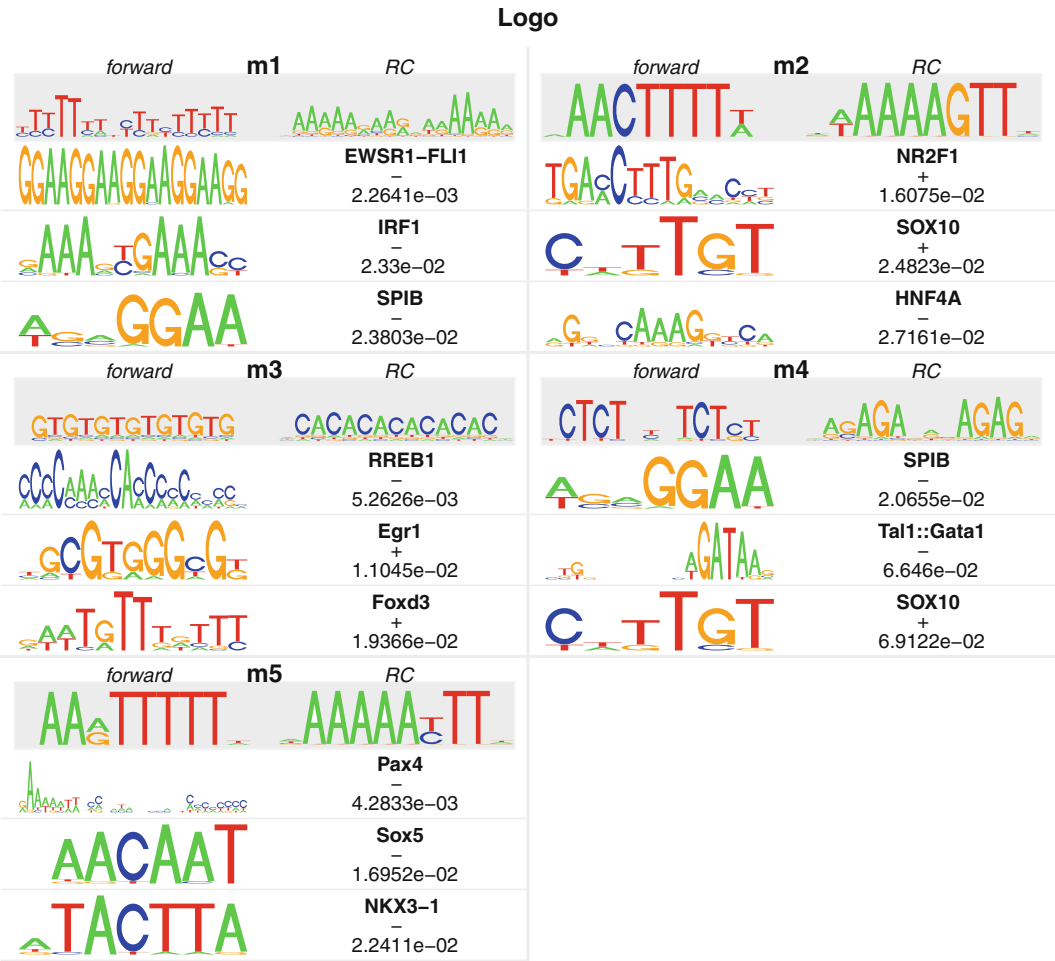


Fig. 8 Identified motifs of DAF-12-binding sites in worm and corresponding matching motifs in JASPAR database. The motifs were generated using rGADEM and the sequences from the putative DAF-12-binding regions merged from three biological replicates identified in worm [32]. The matching motifs in JASPAR database were obtained using MotIV. Five significant motifs are shown. For each discovered motif, both forward and reverse complement logos are given on the *top* and three significant matching known motifs are given below with forward motif logo, motif name, and matching *p*-value

4 Notes

1. DAF-12 binding sites from three ChIP-chip datasets were used as example. The same analysis would apply to ChIP-seq datasets as long as binding sites contain chromosome coordinates.
2. The GNU Scientific Library (GSL) needs to be installed for the MotIV package. GSL is free and can be downloaded at <http://www.gnu.org/software/gsl/> for Unix distributions and at <http://gnuwin32.sourceforge.net/packages/gsl.htm> for Windows. ChIPpeakAnno, rGADEM, MotIV, and rtrack-



Fig. 9 Identified transcription regulatory network of DAF-12 in worm. The network was generated using GeneNetworkBuilder with the data from the putative DAF-12-binding regions merged from three biological replicates identified in worm [32] and expression microarray results from animals lacking DAF-12 [33]. Please note that it would be ideal if the worm stages are similar between the ChIP experiment and expression experiment. For this reason, the tolerance is set to two unverified internal nodes. DAF-12, the root node, is shown as a *square*, the direct targets are depicted as octagons, and all the other nodes are shown as *circles*. The larger the *circle* or *octagon*, the more downstream connections it has. The genes with altered expressions are colored in *yellow*. The genes with binding sites near the promoter region but with unchanged expression are colored in *cyan*. All the miRNAs are colored in *pink*

layer are available from Bioconductor, an open source and open development software project specializing in biological data analysis and integration based on R, a system for statistical computation and graphics [37]. R can be obtained at <http://cran.r-project.org/>. The bioconductor packages can be installed from R by typing `source("http://bioconductor.org/biocLite.R")` and `biocLite(c("ChIPpeakAnno", "rtracklayer", "rGADEM", "MotIV"))`.

3. The parameter *totalTest* indicates how many potential peaks in total are used in the hypergeometric test. It should be larger than the largest number of peaks in the replicates. The smaller it is set, the more stringent the test is. The time used to calculate *p*-value does not depend on the *totalTest*. For practical guidance on how to choose *totalTest*, please refer to the post at <https://stat.ethz.ch/pipermail/bioconductor/2010-November/036540.html>
4. BED format is a common file format that provides a flexible way to define peaks or annotations as data lines detailed at <http://genome.ucsc.edu/FAQ/FAQformat#format1>. For six-field BED format, each line contains six tab-separated fields in the order of chromosome name, chromosome start, chromosome end, line name, score and strand in + or –.
5. For a list of annotation dataset available for other species, use function *listDatasets*.
6. To annotate the peaks with other genomic features, use *getAnnotation* with different *featureType* such as *exon*, *5utr*, and *3utr*. It is also possible to pass customized annotation data into the function *annotatePeakInBatch*, such as a list of binding sites for the same protein from the literature, or a list of binding sites of a different protein functioning as transcription complex together with the protein under study. Prior to calling the function *annotatePeakInBatch*, it is necessary to represent both datasets as *RangedData*, where *start* is the start of the binding site, *end* is the end of the binding site, *names* is the name of the binding site, and *space* and *strand* are the chromosome name and strand, respectively, where the binding site is located.
7. By default, the distance is calculated as the distance between the start of the binding site and the TSS, which is the gene start for genes located on the forward strand and the gene end for genes located on the reverse strand. Users have the flexibility to set *PeakLocForDistance* to *start*, *middle* or *end*, and set *FeatureLocForDistance* to TSS, *start*, *middle*, *end* or *geneEnd*. Also, by default, the function outputs only nearest genes. If interested in including genes overlapping with the peaks in addition to the nearest genes, set *output* = "*both*", *maxgap* = 0 and *select* = "*all*". For customized settings, please type help (*annotatePeakInBatch*) in R.
8. Package *org.Ce.eg.db* needs to be installed in R by typing `source("http://bioconductor.org/biocLite.R")` and `biocLite("org.Ce.eg.db")`.

In addition, not all species-specific annotation packages are Entrez ID centric. For example, *org.At.tair.db* for Arabidopsis is TAIR ID centric and *org.Sc.sgd.db* for Yeast is orf centric. As a

workaround, set *feature_type_id* as *entrez_id* for Arabindopsis and Yeast when using *addGeneIDs* or *getEnrichedGO*.

9. *MaxDistance*, 5,000 bp by default, specifies the maximum gap allowed between the peak start and TSS. In the example, it is set to 1 kb to be more stringent. For additional customized parameter setting, please type *help (peak-sNearBDP)* in R.
10. Only available with ChIPpeakAnno 2.5.9 or later. By default, peaks that reside in within 1 kb upstream (*proximal.promoter.cutoff*) from or overlap with transcription start site are classified as proximal promoters. Peaks that reside within 1,000 bp downstream (*immediate.downstream.cutoff*) of gene end but not overlap 3' UTR are classified as immediate downstream. Peaks that reside over 1,000 bp from gene start or gene end are classified as enhancers. Users have the flexibility to set both parameters different from default value 1,000 in the function *assignChromosomeRegion*.
11. For other organism-specific GO gene mapping packages org.xx.db, please refer to <http://www.bioconductor.org/packages/release/data/annotation/>. The parameter *maxP* is the maximum *p*-value required to be considered significant, *multiAdj* indicates whether to apply multiple hypothesis testing adjustment, *minGOterm* is the minimum count in a genome for a GO term to be included, and *multiAdjMethod* is the multiple testing procedure to be applied. For customized setting, please type in R *help(getEnrichedGO)*. The output contains the corresponding gene entrezIDs for each enriched GO term. Table 2 is simplified with EntrezIDs removed using the following R code.

```
write.table(rbind(unique(enrichedGO$mf[,1:10]),
unique(enrichedGO$bp[,1:10])), file="Table2.xls", sep="\t", row.names=FALSE)
```

12. The function *getEnrichedPATH* is only available with ChIPpeakAnno 2.4.0 or later. Many parameters are the same as for *getEnrichedGO* (see **Note 10**). For customized settings, please type in R *help(getEnrichedPATH)*. The output contains the corresponding gene entrezIDs for each enriched pathway term. Table 3 is simplified with EntrezIDs removed using the following R code.

```
write.table(unique(enrichedPATH[, -2]),
file="Table3.xls", sep="\t", row.names=FALSE)
```

13. Only available in Bioconductor 2.11 or later. To install, type the following two lines in an R session.

```
source("http://bioconductor.org/biocLite.R")
biocLite("BSgenome.Celegans.UCSC.ce10")
```

14. As of today, JASPAR contains very limited motifs for worms. There is an effort going on spearheaded by Bioconductor core team member Paul Shannon to consolidate a richer motif data resource in the R environment.

Acknowledgment

I would like to thank Dr. Michael Brodsky at Program in Gene Function and Expression in University of Massachusetts Medical School for his critical review of the manuscript and his excellent suggestions.

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